JAVA PROGRAMMING PRACTICES

Marks Total and Average Using Command-Line Arguments

Basic Information

Week	Week 3
Source file	MarksCommandLine.java
Concepts	Command-line arguments, loops, arithmetic.

Objective / Aim

To implement marks total and average using command-line arguments using the concepts listed in the R23 Java Programming Practices syllabus.

System Requirements

JDK (Java Development Kit) and a text editor or Java IDE.

Algorithm / Logic Description

1. Check that five command-line arguments are supplied.
2. Convert the arguments to integers and calculate total.
3. Calculate and display the average.

UML / Program Structure

```
MarksCommandLine
main() demonstrates the algorithm above.
```

Source Code

```
public class MarksCommandLine
{
    public static void main(String[] args)
    {
        if (args.length != 5)
        {
            System.out.println("Usage: java MarksCommandLine mark1 mark2 mark3 mark4 mark5");
            return;
        }
        int total = 0;
        for (int i = 0; i < 5; i++) total += Integer.parseInt(args[i]);
        System.out.println("Total: " + total);
        System.out.println("Average: " + total / 5.0);
    }
}
```

Compilation and Execution

```
javac MarksCommandLine.java
java MarksCommandLine 85 90 78 88 94
```

Output Screen

```
Total: 435
Average: 87.0
```

Result

The program demonstrates the listed syllabus concepts. The sample output is based on a compiled execution.